

AI Pricing in the Macroeconomy

Predictions for the Phillips Curve and Monetary Policy

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The views expressed herein are those of the author and do not necessarily represent the views of the Federal Reserve Bank of Kansas City or the Federal Reserve System.

Motivation

Some Motivating Evidence

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- Theory says the FPA matters for monetary policy

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- A maximally tractable approach: similar to a menu cost model, so we can use existing results
- Characterize how AI pricing changes the Phillips curve and monetary policy
- Provide 4 predictions for future AI pricing + monetary policy

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4. Optimal monetary policy will respond more aggressively to inflation

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- Richer models are more costly
- But are more valuable for decision-making

What Is AI Pricing? (Example)

PLACEHOLDER

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- Firm's model complexity = how many factors $F_{ij}(k)$ to use

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 - $\implies$ different behavior

Gap Representation

- *Gross gap* is factor k -specific:

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- Integrating over factors k and products j , the firm's instantaneous loss is

$$\mathcal{L}_{it} = \underbrace{-\mathbf{B}M_i \int_0^{A_i} g_{it}(k)^2 dk}_{\text{modeled factors}} - \overbrace{\mathbf{B}M_i(1 - A_i)V_{\hat{z}}}^{\text{unmodeled factors}}$$

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 - Looks like a menu cost model, but for a single factor k applied to M_i markets

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 3. demand is more elastic
- We see (1) and (2) in the data (Adams et al., 2026)
- Similar predictions for *frequency of price adjustment* (ongoing work)

An AI Phillips Curve?

- Modeled factors $k \in [0, A_i]$: Calvo-plus menu cost dynamics

Aggregation

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 - The *Primary Eigenvalue Decomposition* reduces the infinite dimensional state
 - Special case with zero trend inflation: isomorphic to Calvo

The AI Phillips Curve

Theorem: the *Primary Eigenvalue Decomposition* implies

$$\pi_t = \Lambda(A_i) mc_t + \beta \mathbb{E}_t \pi_{t+1}$$

with slope

$$\Lambda(A_i) \equiv \Omega \left(A_i \frac{(1 - \theta_{A_i})(1 - \beta \theta_{A_i})}{\theta_{A_i}} + (1 - A_i) \frac{(1 - \theta_{\neg A_i})(1 - \beta \theta_{\neg A_i})}{\theta_{\neg A_i}} \right)$$

$$\theta_{A_i} \equiv \exp \left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2} \right) \quad \theta_{\neg A_i} \equiv \exp(-\zeta)$$

► PED details

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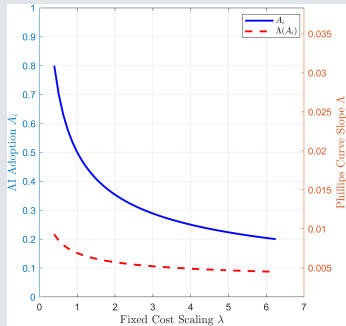
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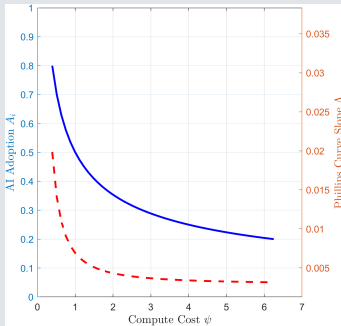
Slope $\Lambda(A_i)$ is increasing in A_i !

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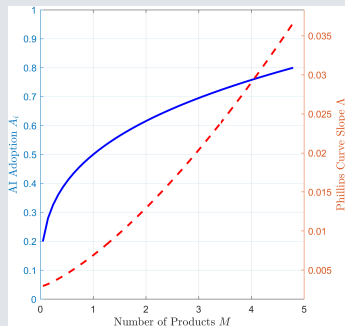
How Do AI Determinants Change the Slope?



(a) Fixed cost



(b) Compute cost



(c) Firm size

AI-NK Model

An AI-Pricing New Keynesian Model

- Replace the pricing block in a textbook New Keynesian model:

AI Phillips Curve: $\pi_t = \Lambda(A_i)\omega y_t + \beta \mathbb{E}_t \pi_{t+1} + u_t^c$

Euler equation: $\gamma y_t = \mathbb{E}_t[\gamma y_{t+1} - (i_t - \mathbb{E}_t \pi_{t+1})] + u_t^d$

Taylor rule: $i_t = \phi_\pi \pi_t + \phi_y y_t + u_t^m$

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- Explore how A_i rising affects the macroeconomy

Calibration

Object	Value	Source
Discount rate ρ	0.04	Standard annual rate
Elasticity of substitution η	6	Standard
Strategic complementarity Ω	0.55	Gagliardone et al. (2025)
Taylor Rule ϕ_π, ϕ_y	1.5, 0.5	Standard
Baseline AI adoption $\bar{A}$	0.5	Normalization
Diffusion variance σ_β^2	0.0073	Match moments
Poisson reset rate ζ	0.85	Match moments
Inaction boundary $\bar{x}$	0.88	Match moments
FPA F	1.26	French CPI microdata (Alvarez et al., 2024)
Std. dev. of price changes	0.076	French CPI microdata
Kurtosis of price changes	5.05	French CPI microdata

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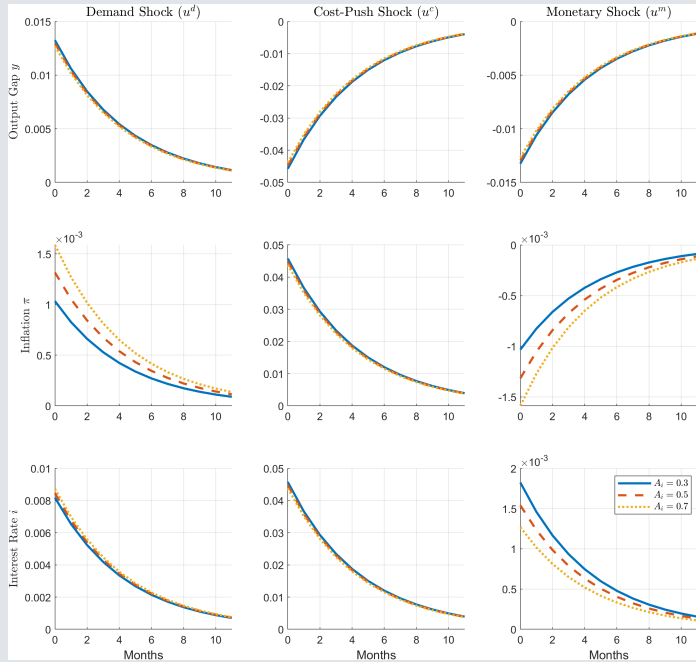
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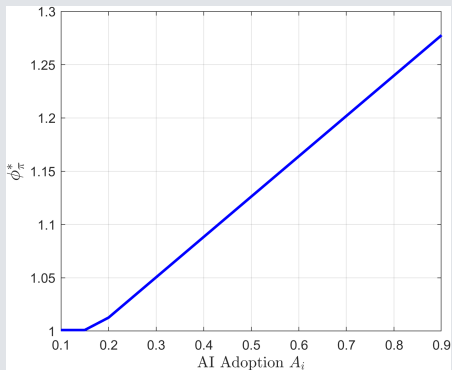
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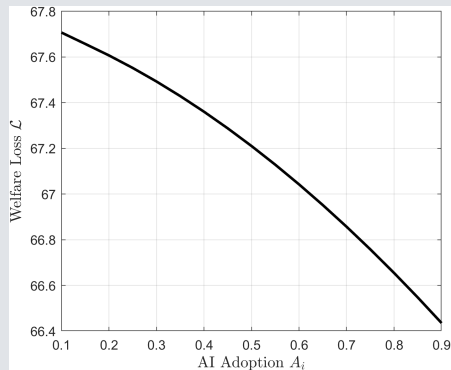
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- little change to cost-push shocks (enter the Phillips curve directly)
- Standard: changing the Phillips curve slope changes optimal monetary policy



Optimal Monetary Policy



(a) Optimal inflation response ϕ_π^*



(b) Minimized welfare loss $\mathcal{W}$

More AI pricing $\implies$ **more aggressive** monetary policy!

What's Next?

- The predictions so far:
 1. AI pricing will continue to rise as costs decline
 2. Prices will become more flexible
 3. The Phillips curve will become steeper
 4. Optimal monetary policy will respond more aggressively to inflation
- Gathering evidence relating AI use + pricing statistics
- New pricing theories!
- Really need better discipline on the AI pricing cost function (help!)

References

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- Cavallo, Alberto**, “More Amazon effects: online competition and pricing

PED Mapping

- For each modeled factor, zero-trend-inflation dynamics are approximated by:

$$x_{k,t}^* = (1 - \beta\theta_k)a_t + \beta\theta_k\mathbb{E}_t x_{k,t+1}^*$$

$$x_{k,t} = (1 - \theta_k)x_{k,t}^* + \theta_k x_{k,t-1}$$

- Modeled and unmodeled factors differ only in effective stickiness:

$$\theta_{A_i} = \exp\left(-\zeta - \frac{\sigma_\beta^2 2\pi^2}{\ell^2}\right), \quad \theta_{\neg A_i} = \exp(-\zeta)$$

Price Change Moments

- Economy-wide price changes are a mixture:

$$F = A_i F^{CP} + (1 - A_i) F^C$$

$$\sigma_{\Delta p}^2 = \frac{A_i F^{CP} (\sigma_{\Delta p}^{CP})^2 + (1 - A_i) F^C (\sigma_{\Delta p}^C)^2}{F}$$

- Targets: frequency, standard deviation, kurtosis
- Baseline pins down $(\sigma_{\beta}^2, \zeta, \bar{x})$